

LAB 6: Set up ECS (Elastic Container Service) by setting up Nginx on ECS.

Lab Report Submitted to the

VIT BHOPAL UNIVERSITY

BACHELOR OF TECHNOLOGY

COMPUTER SCIENCE ENGINEERING



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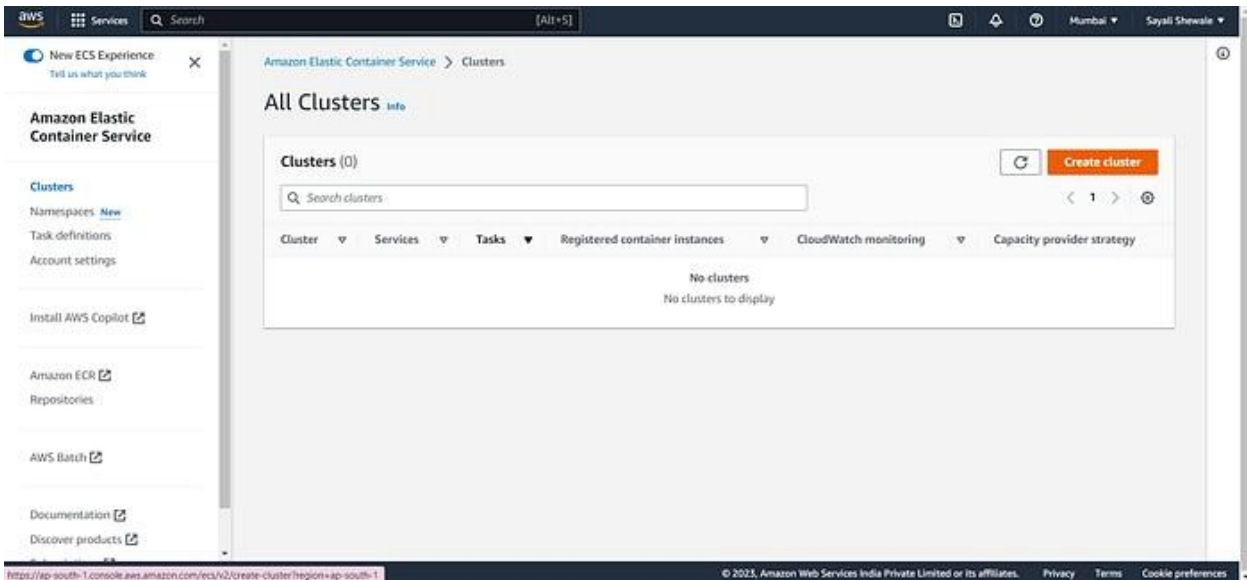
VIT Bhopal University, Madhya Pradesh

Feb, 2026

AIM: Set up ECS (Elastic Container Service) by setting up Nginx on ECS.

1. Create an ECS cluster:

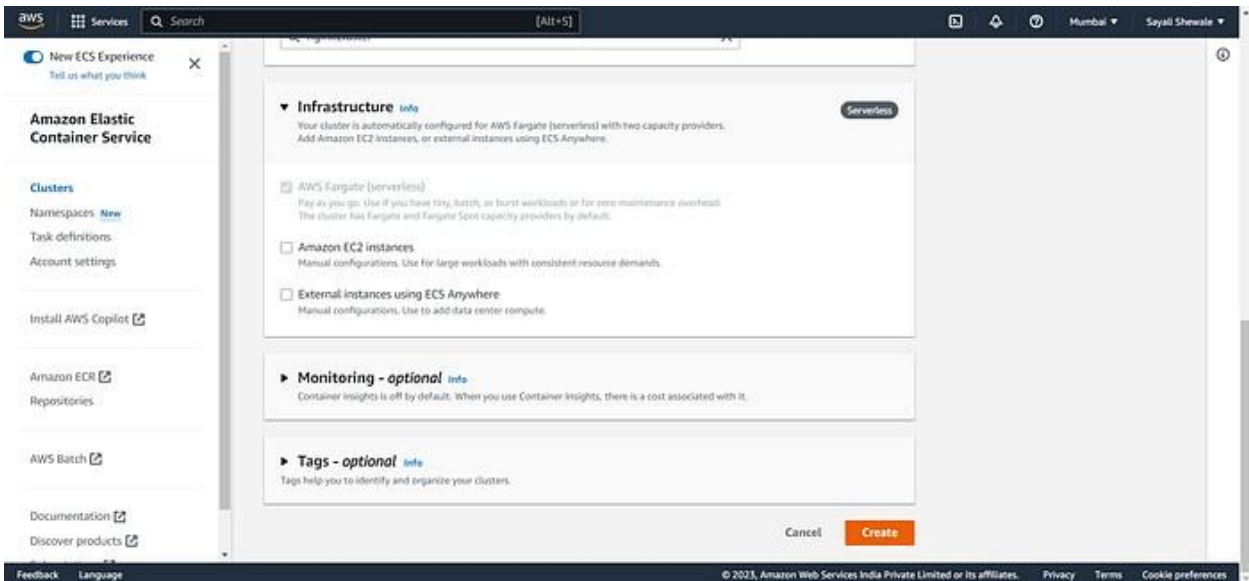
In the ECS service, click on “Create Cluster”.



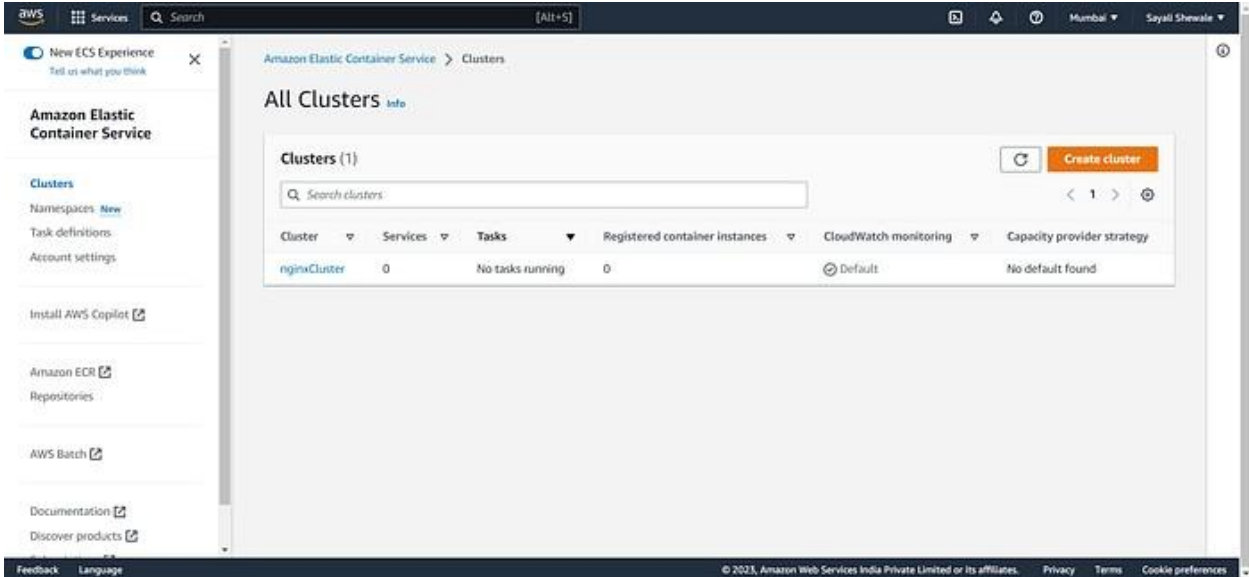
Configure the cluster settings, such as the cluster name, VPC, subnet.



Click on 'Create'

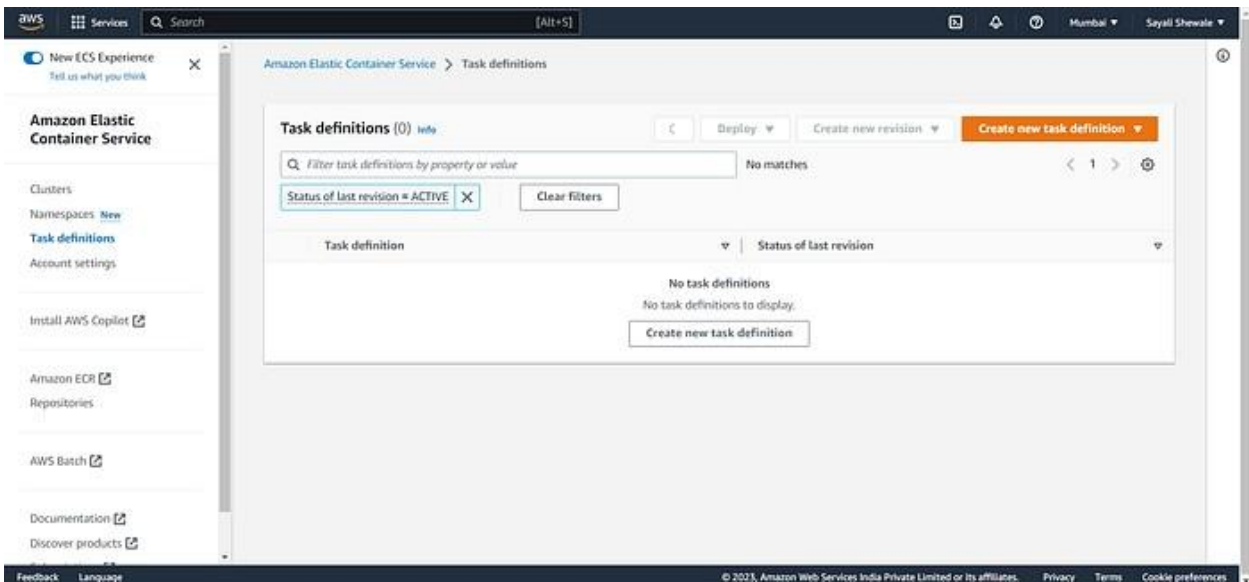


Cluster is successfully created.

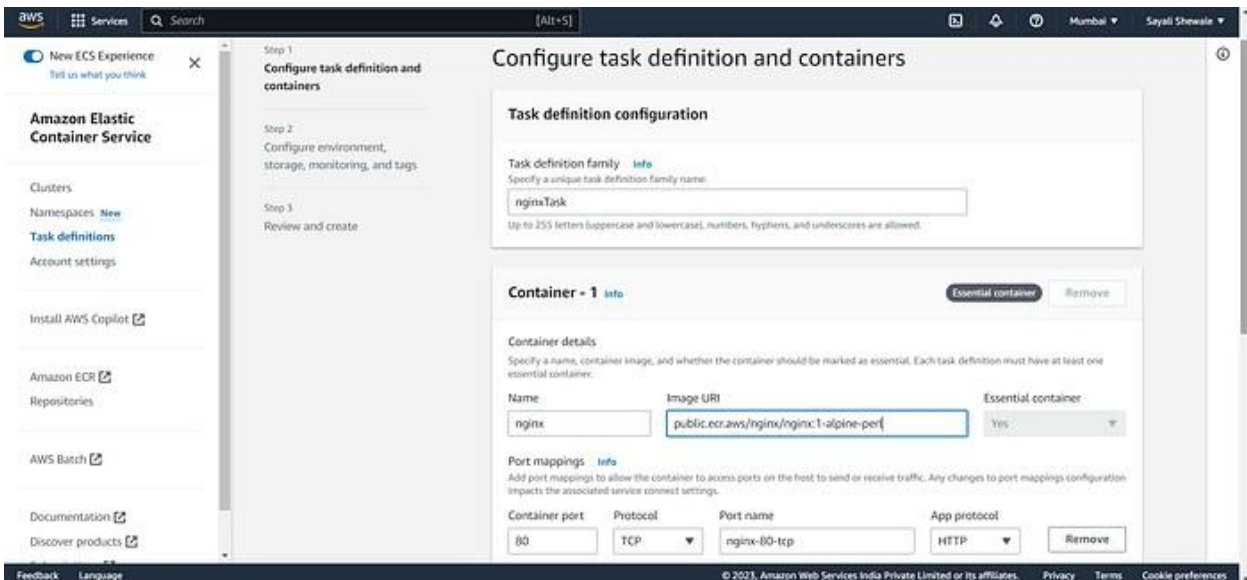


2. Create a task definition:

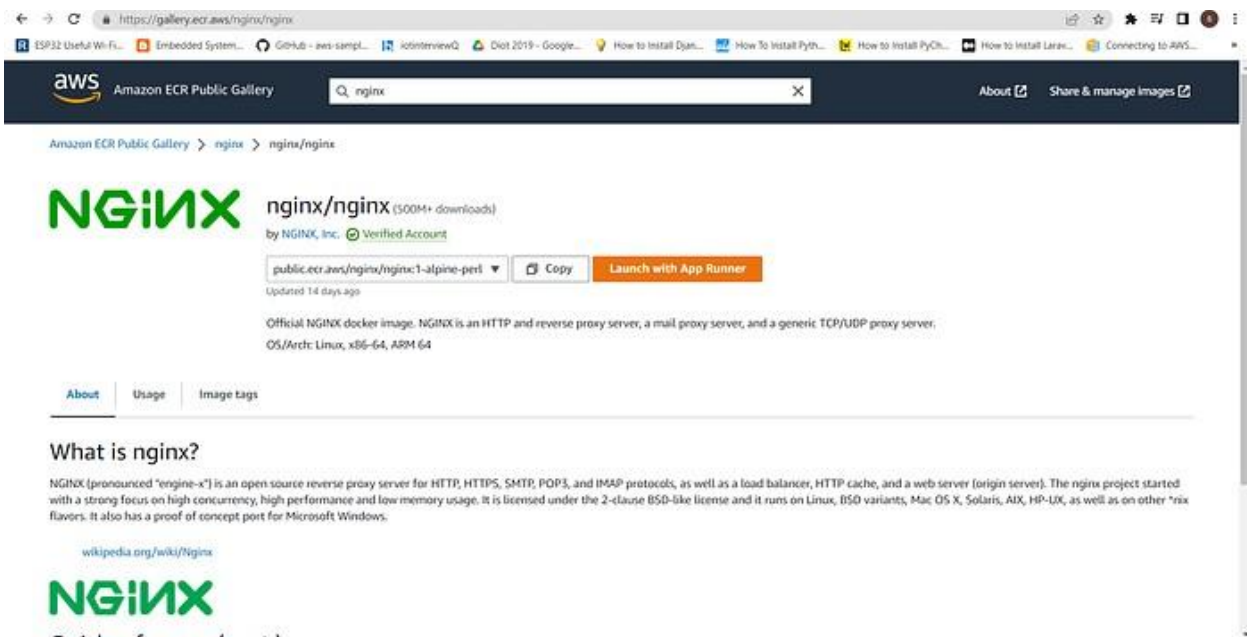
In the ECS service, click on “Task Definitions”, and then click on “Create new Task Definition”.



Set the container name to “nginx”, copy and paste Image url from ‘Amazon ECR public gallery’ site. Specify the port mappings for HTTP , by setting the “Container port” to 80.



Browse Amazon ECR public gallery and search for nginx image and copy that image url.



Click on 'Next'

This screenshot shows the 'Add more containers' step in the AWS ECS console. The left sidebar contains the 'Amazon Elastic Container Service' navigation menu. The main panel is titled 'Add more containers' and includes a table for 'Container port', 'Protocol', 'Port name', and 'App protocol'. A single mapping is shown: Container port 80, Protocol TCP, Port name nginx-80-tcp, and App protocol HTTP. Below the table are sections for 'Environment variables - optional', 'Add from file', 'HealthCheck - optional', and 'Docker configuration - optional'. At the bottom right, there are 'Cancel' and 'Next' buttons.

Container port	Protocol	Port name	App protocol
80	TCP	nginx-80-tcp	HTTP

Buttons: Add more port mappings, Add environment variable, Add environment file, Add more containers, Cancel, Next.

Choose the Fargate launch type, Configure the task settings, such as the task memory and CPU limits, here CPU is .5vCPU and memory is 1GB.

This screenshot shows the 'Configure environment, storage, monitoring, and tags' step in the AWS ECS console. The left sidebar is the same as the previous screenshot. The main panel is titled 'Configure environment, storage, monitoring, and tags' and shows the 'Environment' section. It includes a dropdown for 'App environment' (set to 'AWS Fargate (serverless)'), a dropdown for 'Operating system/Architecture' (set to 'Linux/X86_64'), and a 'Task size' section with 'CPU' set to '.5 vCPU' and 'Memory' set to '1 GB'. There are also sections for 'Container size - optional' and 'Task roles, network mode - conditional'. At the bottom right, there are 'None' and 'Next' buttons.

Environment configuration:

- App environment: AWS Fargate (serverless)
- Operating system/Architecture: Linux/X86_64
- Task size: CPU .5 vCPU, Memory 1 GB

Buttons: None, Next.

Key **Value type** **Value**

awslogs-group	Value	/ecs/nginxTask
awslogs-region	Value	ap-south-1
awslogs-stream-prefix	Value	ecs
awslogs-create-group	Value	true

[Add](#) [Remove](#)

☐ **Use trace collection** [Info](#)
 Amazon ECS creates an AWS Distro for OpenTelemetry sidecar to route traces from your application to AWS X-Ray. See pricing information on AWS X-Ray [Info](#)

☐ **Use metric collection** [Info](#) [Preview](#)
 Amazon ECS creates an AWS Distro for OpenTelemetry sidecar to route custom container and application metrics to Amazon CloudWatch or Amazon Managed Service for Prometheus.

► **Tags - optional** [Info](#)
 Tags help you to identify and organize your task definitions.

[Cancel](#) [Previous](#) [Next](#)

Click on 'Create'

Container log collection

Destination
 Amazon CloudWatch

awslogs-group	Value	/ecs/nginxTask
awslogs-region	Value	ap-south-1
awslogs-stream-prefix	Value	ecs
awslogs-create-group	Value	true

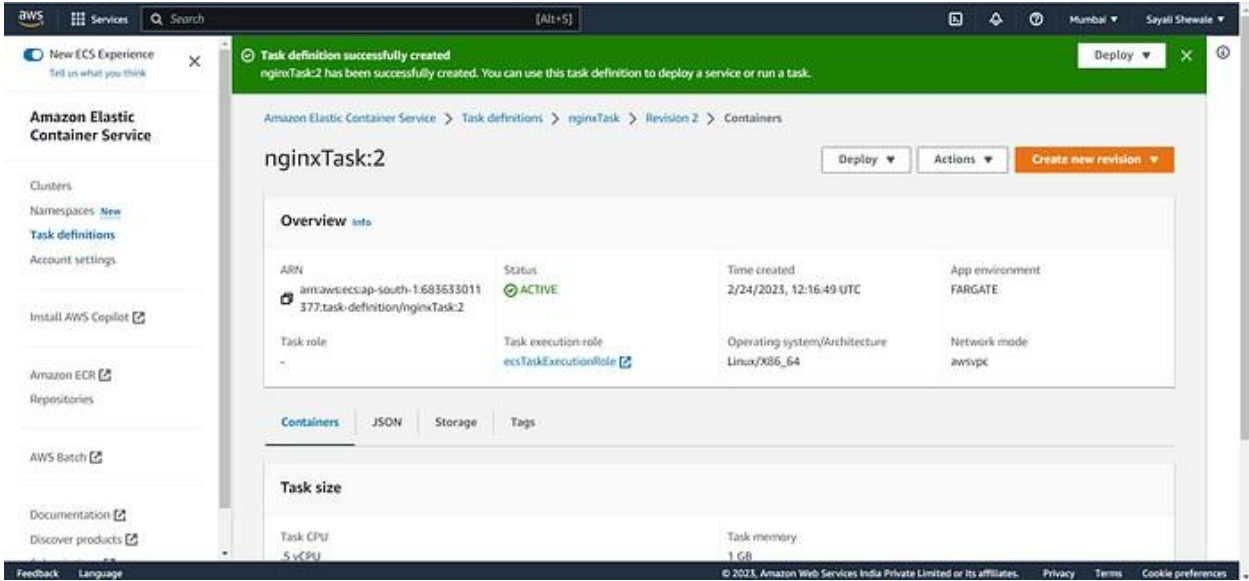
Trace collection
 Off

Metric collection
 Off

Tags
 No tags added

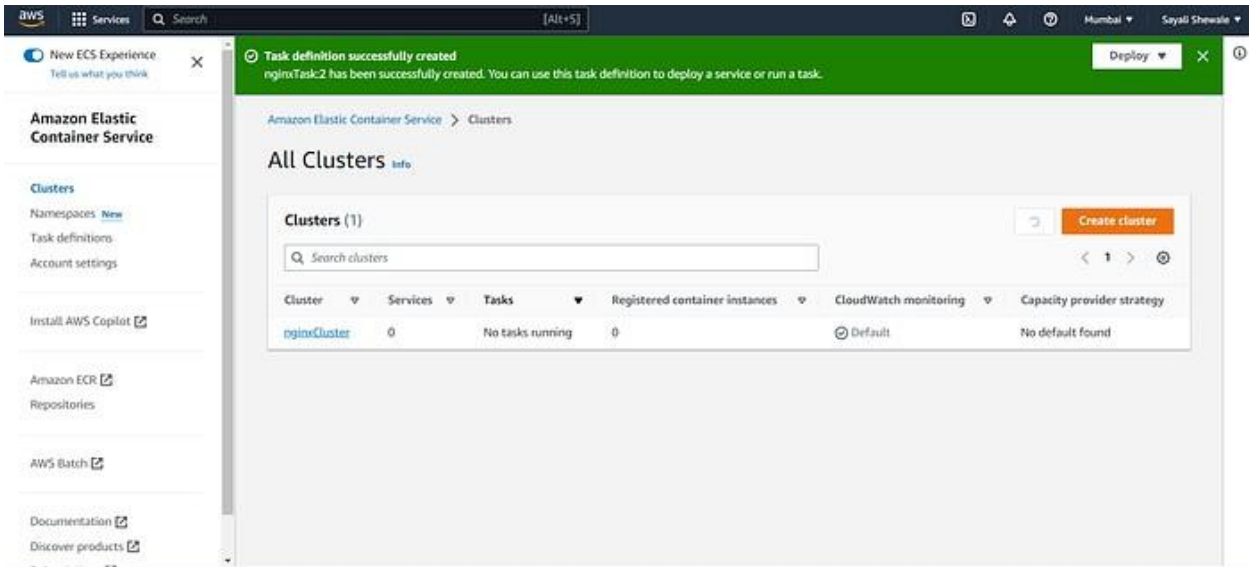
[Cancel](#) [Previous](#) [Create](#)

Task is successfully created.

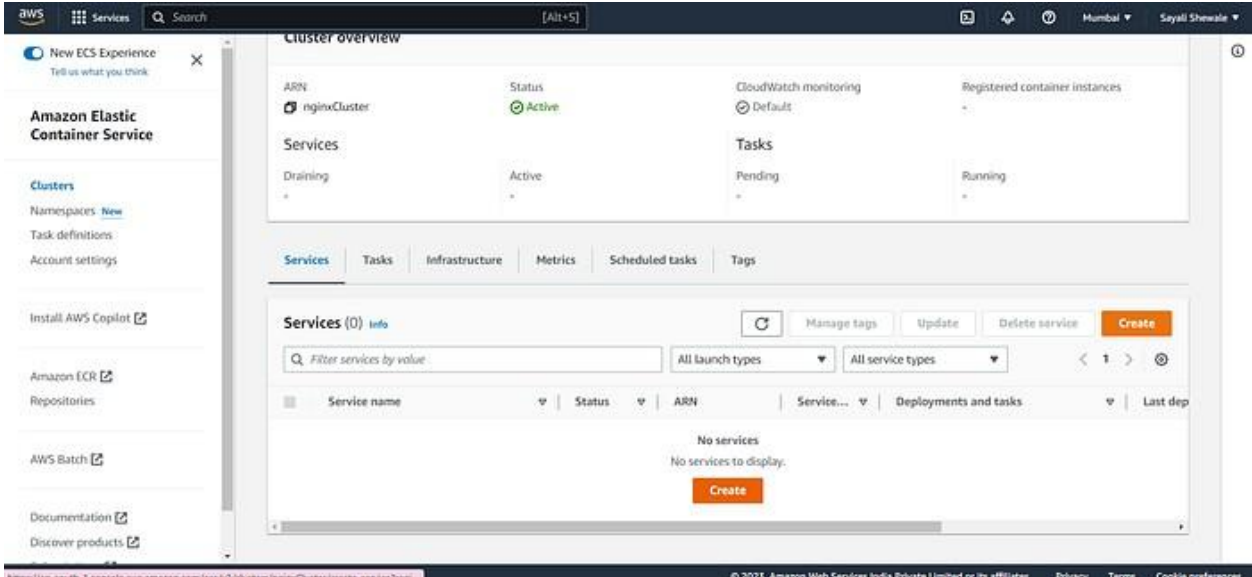


3. Create a service:

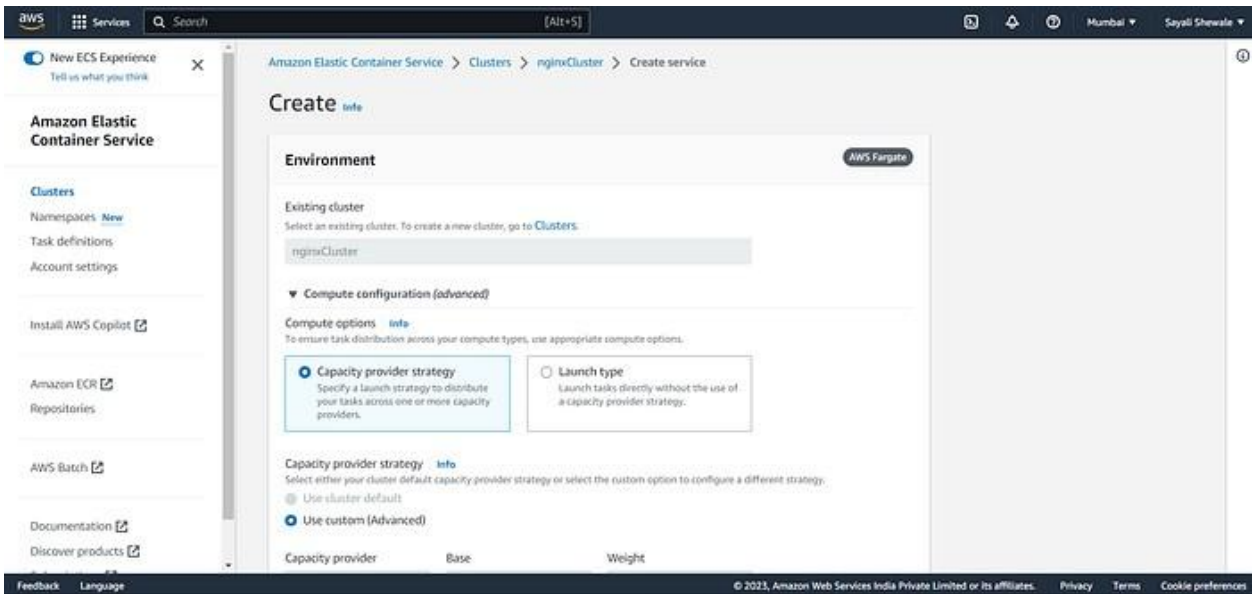
In the ECS, click on “Clusters”, and select the cluster that you created.

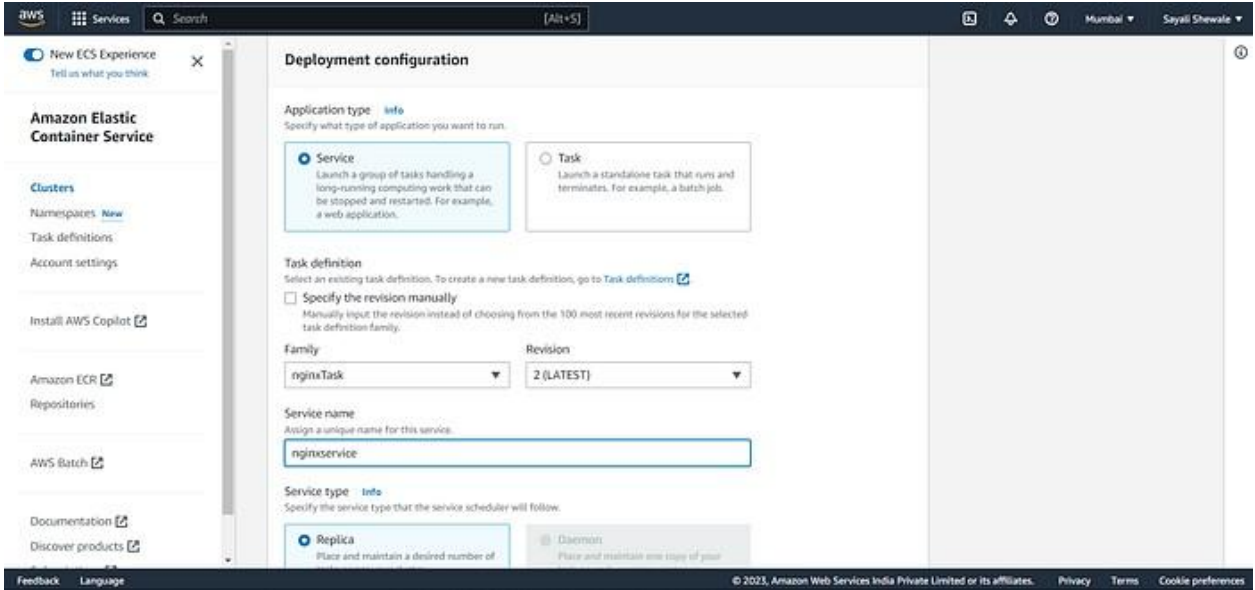


Click on “Create Service.

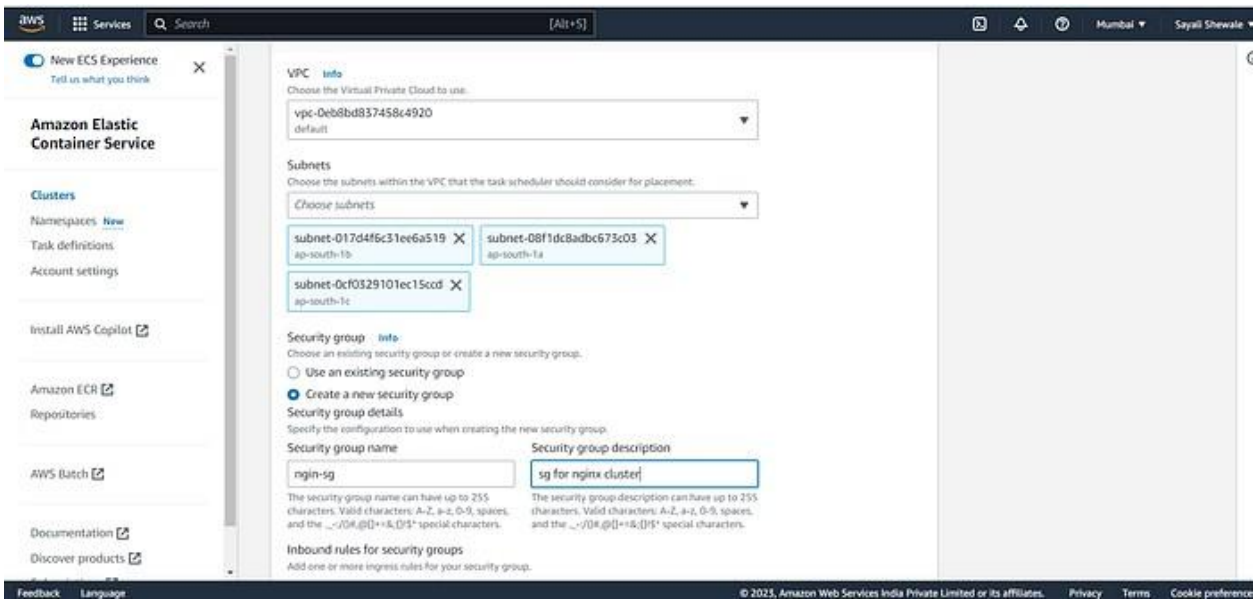


Choose the task definition that you created in above steps.

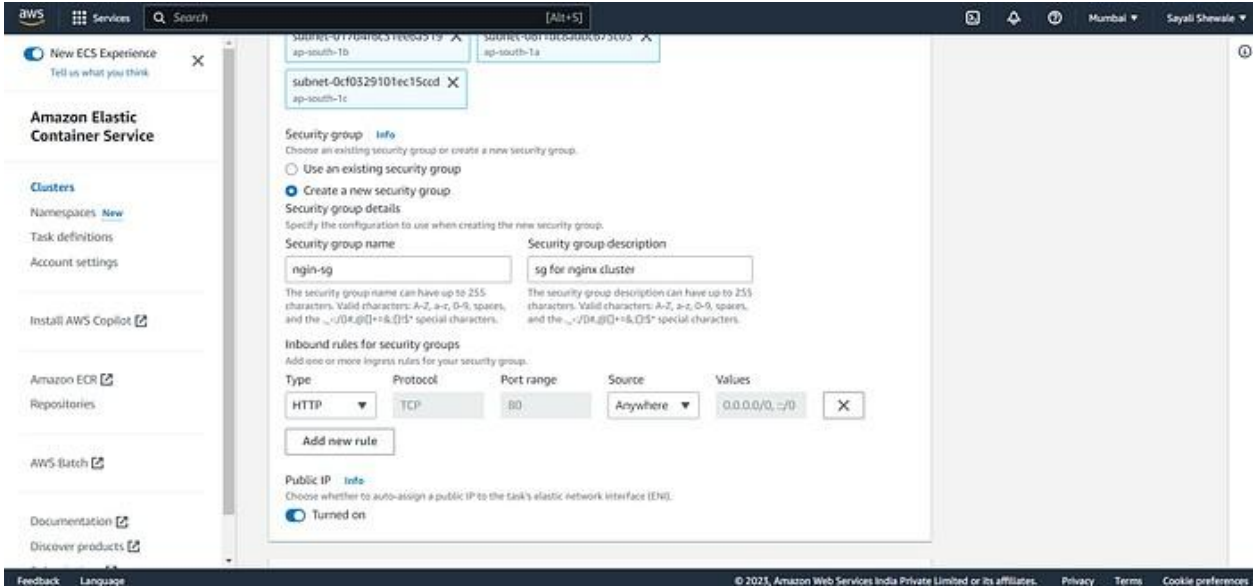




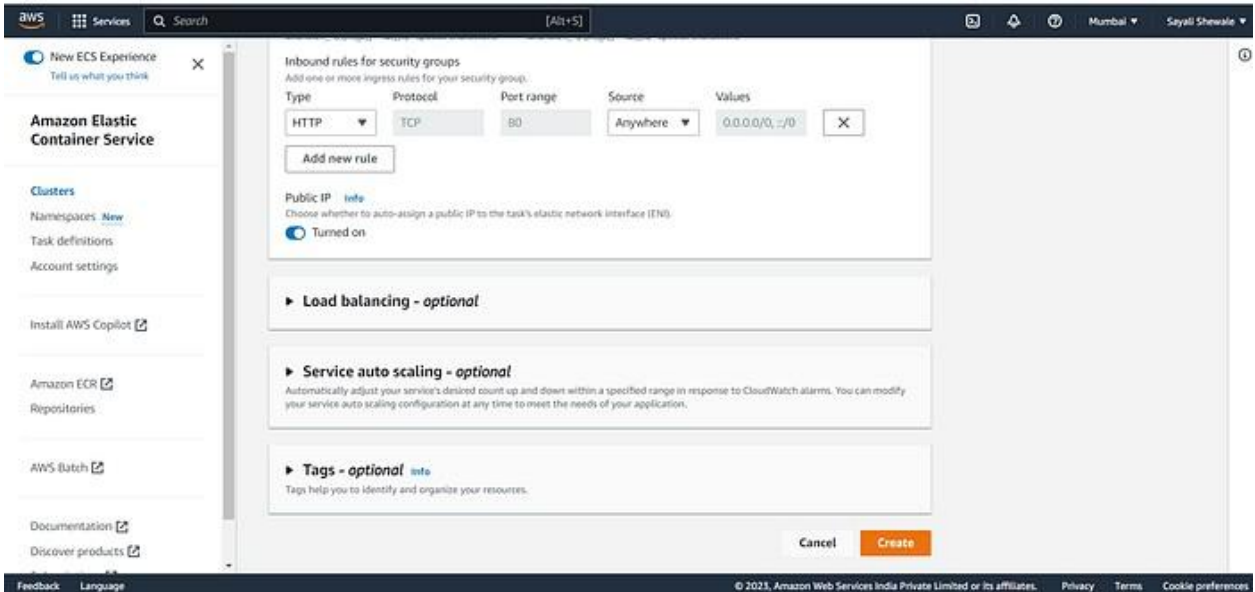
configure the service settings, such as the VPC, subnet, and security group settings. Create a new security group



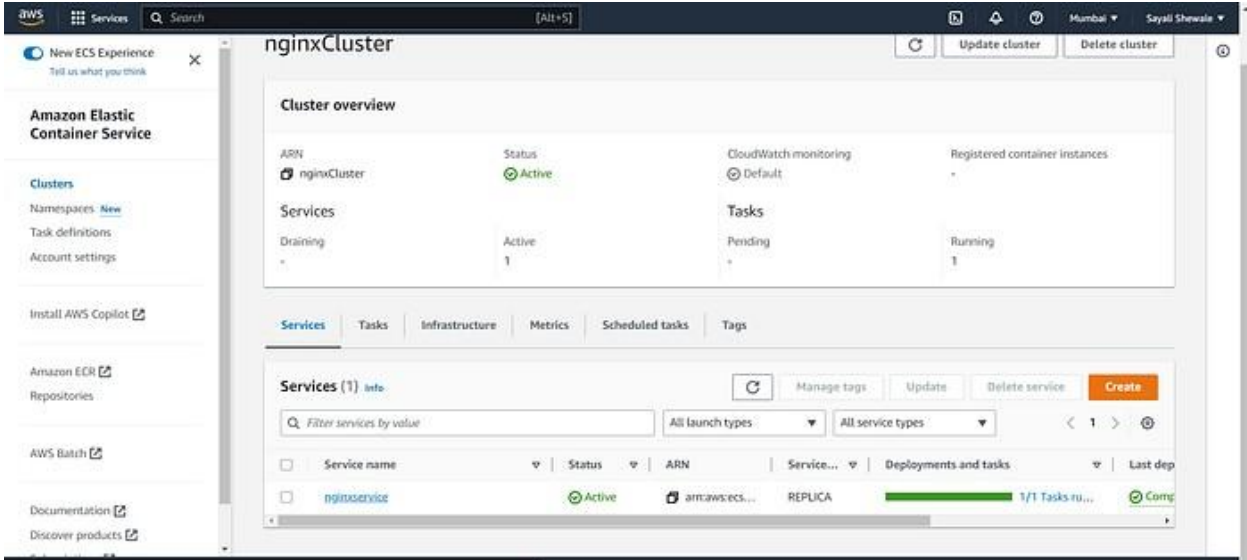
Specify the port mappings for HTTP with port 80.



Click on 'Create'.



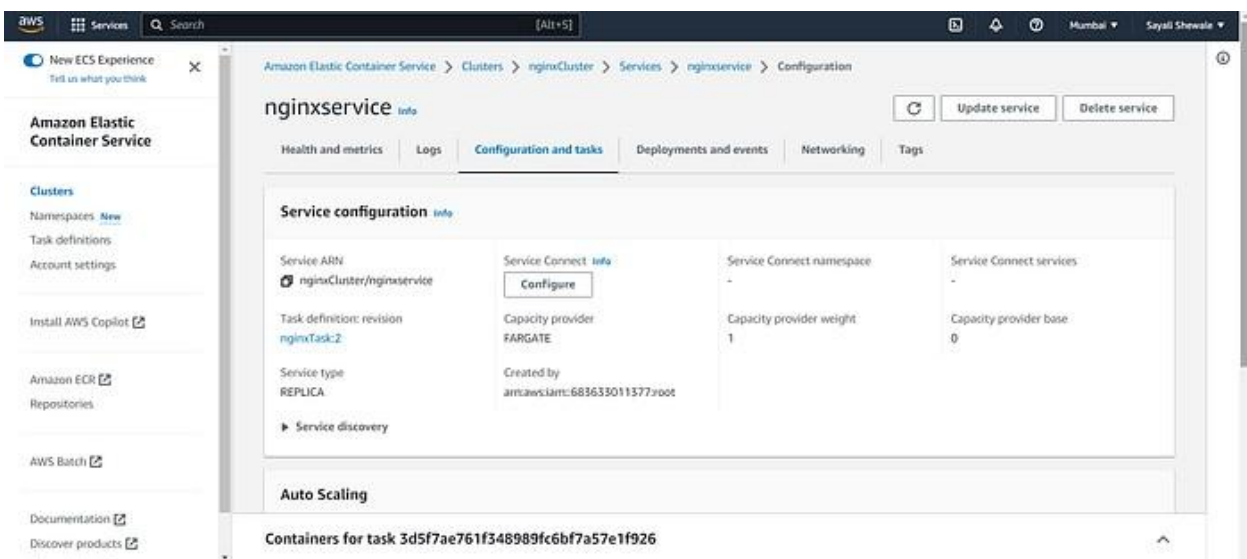
Service is successfully created.



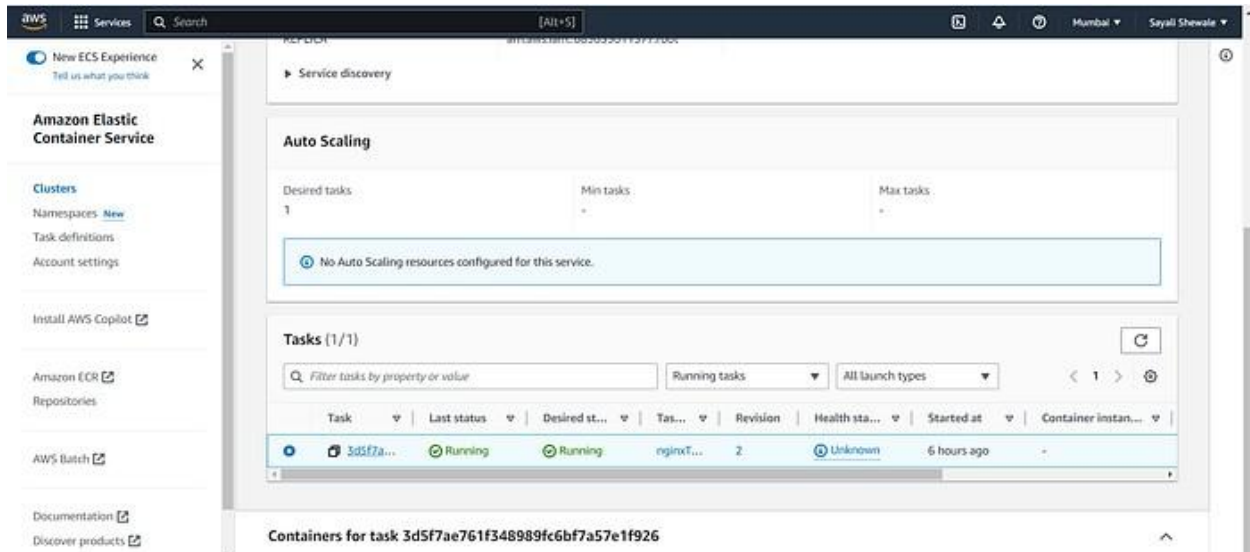
Once the ECS service is running, test the Nginx container by accessing the public IP address of the Fargate task in a web browser. You can find the public IP address in the “Tasks” tab of the ECS service, under the “Configuration” section.

4. Test the Nginx container:

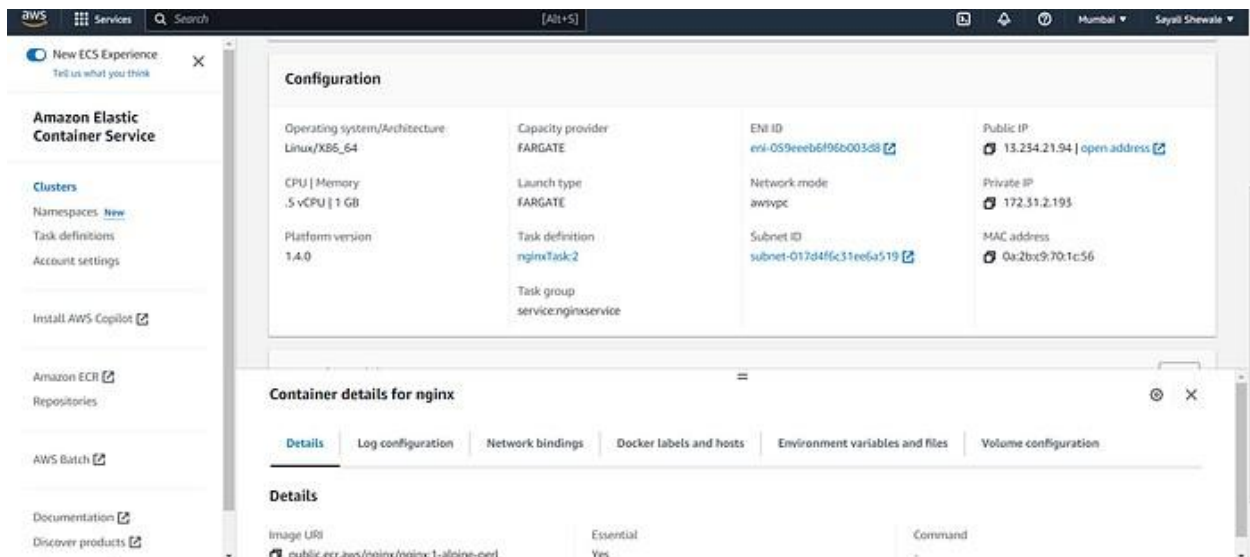
Click on ‘Cluster’ that you created.



Click on 'Task'.



In task, go to 'configuration' section, there you can find public ip.



Browse Public IP address, you can see the default Nginx welcome page.

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.